GAME1SRC (EDM2)

1 Motivation

Fast graphics with DIVE.

A port from C to fpc Pascal (FPC 3.2.2).

2 C code

2.1 Background

The program shows a few drawing primitives (2 black squares (outlines) and a red one in the center) on a white background.

Original source is published here: https://www.edm2.com/index.php/Gearing_Up_For_Games

2.2 Pointer arithmetic

See the code. Used a lot here.

2.3 Array dimension for image

See the site/pdf:

> *DIVE graphics work around a buffer. The buffer is simply a block of memory that is organized as a two dimensional array for the x and y coordinates. All the pixels in a horizontal stretch are stored one after another to form a row, and the rows are stored one after the other with one row per column.*

A row is a horizontal line, a column a vertical one.

2.4 Arrays (color)

No remarks.

2.5 Arrays (data)

No remarks.

2.6 Color handling

Done using predefined constants.

2.7 Compilation

Tried to compile with Watcom C++ 1.9, as follows:

- TBD

Resources:

- change the .RC file to include GAME1.ICO
- next, rc -r GAME1.RC

2.8 Running the program

No problem :).

3 Pascal code (fpc)

3.1 Variable names

Pascal is not case-sensitive so I inserted an underscore after h. Example: hab becomes h_ab.

3.2 Pointer arithmetic

See the code. Used a lot here. For this program it is simpler to change apbyRowStart into a **dynamic** array of PBYTE. But a fixed array could also do.

3.3 Array dimension for image

As in the C code.

3.4 Color handling

Functional after changes in dive unit.

3.5 Other

I “have” to disable

```
WinLoadString  (h_ab, 0, ID_APPMAIN, sizeof(szTitle), szTitle);
```

Apparently one can do without…

3.6 Compilation

Resources:

- change the .RC file to include GAME1.ICO
- next, rc -r GAME1.RC

And in source: {$R GAME1.RES}

3.7 Running the program

No problem :).

3.8 Unit issues: pmwin, dive

Have latest !

[20261103] Unit dive required more changes (from game1src):

- in Functions DiveSetDestinationPalette, DiveSetSourcePalette: replace last arg
  - Var bRGB2Entries : Pointer)
- by (remove var)
  - bRGB2Entries : Pointer)

3.9 Quality

Program can be compiled with –Cr -gh -gl; the result runs.

4 Revisions

[20261106] public version